

Thesis for the Master of *Science*

*Exploring Factors
Influence Human Behavior of AI Personality
in a Virtual Collaborative Environment*

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Graduate School of Hanyang University

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Thesis for Master of Science

*Exploring Factors Influencing Human Behavior of AI Personality
in a Virtual Collaboration Environments*

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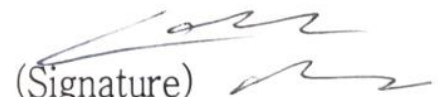
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ABSTRACT

Exploring factors Influence Human Behavior of AI Personality in a Virtual Collaborative Environment

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In the evolving landscape of artificial intelligence (AI) and robotics, technology has transitioned from mere auxiliary tools to central collaborative partners in human activities, fostering cooperative interactions. AI's consistent behavioral patterns and characteristics have the potential to reduce uncertainty in real-world interactions. However, the predominant focus has been on an "explicit" approach, deliberately manipulating AI features, such as word choice, speech pace, and facial expressions. This approach often leads to predefined interactions, but it does not consider the human tendencies to anthropomorphize AI, potentially causing conflicts between AI's intended personalities and user perceptions. Therefore, this study investigates factors shaping human behavioral changes concerning AI personalities through experiments conducted in a virtual environment where AI agents with distinct personalities collaborate for survival. The findings from these experiments illuminate important considerations for future research on AI personalities. Data on the behavior,

speech, and surveys of 36 Korean participants engaged in real-time voice interactions with AI agents in the 'Don't Starve Together' game were collected and compared for statistically significant differences between experimental groups. The results indicated significant intergroup differences in participant movement entropy. However, no significant differences were observed in speech frequency and survey responses based on the AI agent. Future research may explore the development of models predicting AI personalities and interactions using reinforcement learning and generative models to facilitate natural and non-mechanical interactions.

Chapter 1. Introduction

In the evolving landscape of artificial intelligence and robotics engineering, artificial intelligence agents and robots are transitioning from mere auxiliary tools to central collaborative partners in human activities [1]. Particularly with the emergence of Large Language Models (LLMs), the interaction between users and AI or robots through language has undergone significant transformation over the past decade [2]. However, current approaches to developing conversational agents often focus on superficially mimicking human traits, such as manipulating speech speed or tone to artificially impart personalities to robots and guiding human perceptions through predetermined interactions [3, 4].

This approach involves developers intentionally manipulating features of AI, such as word choice, speech pace, facial expressions, and more, with the intention of eliciting specific positive responses, often referred to as an “explicit” approach. For instance, in the case of ‘Character AI,’ a virtual chatbot service platform based on GPT-3, users have the capability to create characters, define their personalities, set specific parameters, and then share them with the community for interaction [5]. Similarly, ‘AI Ruda,’ introduced as an AI Companion, embodies the persona of a vibrant 21-year-old female college student majoring in psychology in Korea and is offered as a virtual chatbot service [6]. It is akin to how specific characters are created in novels or movie scripts according to the author’s imagination [7]. In such an approach, without the reader’s intervention, the character’s personality is pre-defined solely based on the author’s intentions and within the confines of the scenario. The character’s

personality is conveyed to the audience, convincingly structured through specific events and coherence. Here, the reader interprets and accepts the author's intentions, and ideas beyond the text and context fall within the realms of interpretation and misinterpretation. This holds true even in explicit programming, where a series of pre-defined if-else exceptions control uncertainty, ensuring that the program flows as intended by the developer. If the user causes the program to operate differently from the developer's intent, errors may occur.

However, in the real world of interacting with AI, not all scenarios align with the developer's expectations and anticipations [8]. While it's possible to base scenarios on ideal conditions, in uncontrolled environments, developers often encounter unforeseen situations, and the outcomes of interactions become unpredictable. Despite reports of measuring intrinsic and consistent personalities for LLMs in personality tests [9], intelligent conversational agents that interact with users exhibit non-homogeneous dialog sessions. Consequently, they can provide conflicting responses to the same question, and many interactions are not reproducible. In other words, in such scenarios, achieving consistent and reliable results tailored to individual users based on pre-defined scenarios becomes challenging.

This is because, unlike in movies, in situations involving social interactions, users are not passive readers who unilaterally follow the creator's intentions; they actively participate in and contribute to the scenario. In such cases, it is nearly impossible for developers to predict all possible scenarios. Moreover, in social interactions, the same signals or information may need to be interpreted differently depending on the context [10].

Overlooking this tendency of humans and instead viewing the entity as devoid of intentions or desires reduces it to an object bound by internal or external rules, not truly engaging in meaningful interaction. This is because social responses are not merely about explicit outcomes but also about the underlying goal-oriented intentions in a social context, where the same result can be interpreted differently. In short, in natural intelligence there is something we want, and the social interaction is a form of “negotiation” to achieve that goal [11].

Approaches to address these complex issues, using methods such as machine learning, deep learning, and reinforcement learning, have been proposed and have generated significant impact [12]. In fact, in some tasks, reinforcement learning has allowed agents to achieve superior performance compared to humans and exhibit more human-like behaviors [13].

Humans, too, acquire appropriate social responses through repetitive learning and use them as the basis for establishing social relationships with others. The book “Social Learning and Imitation” provides a comprehensive exploration of social learning and imitation as fundamental processes through which individuals observe and interact with others, acquiring new behaviors, beliefs, and attitudes [14]. AI, similarly, may be able to implement human-like traits by mimicking such mechanisms of pattern recognition in vast datasets. However, it is a separate challenge to build systems that can recognize human personality traits. This is because humans tend to attribute personality traits to entities, even those that do not possess inherent personalities.

This phenomenon is often referred to as the “Eliza effect,” where even the most rudimentary forms of artificial intelligence led humans to unconsciously anthropomorphize them [15]. Although Eliza merely parroted

back what patients said, people believed she possessed personality. While entrusting AI with tasks like autonomous driving requires capabilities equal to or surpassing those of humans, the goal for AI or robots in terms of personality may not be limited solely to “perfection” or “precision.” Furthermore, in the context of social interactions with humans, the expression of AI’s personality requires it to become an entity with some form of independent consciousness. Even below the level of human-level implementation, if AI can induce the perception of personality in the eyes of humans when they interact with it, it can have an immediate impact on human behavior, preferences, and trust in AI [16]. It can also become an influential aspect of Social AI, evolving to become a suitable form of companion for individuals [17].

However, as mentioned earlier, if personality is explicitly defined without a deep understanding of human behavior, there may be a mismatch between the intended personality direction by the creator and the way it is perceived by humans [18]. Even for the same agent, individuals may perceive and accept its personality differently. This not only makes it difficult to implement consistent personalities but also increases uncertainty about the direction of interactions with humans, potentially undermining overall system trustworthiness. For example, if a creator intends to implement a helpful AI, but users perceive it as having an overbearing personality, they may adopt a defensive attitude and avoid using the AI, resulting in unintended negative consequences.

Therefore, despite the emphasis on advanced AI technologies and generative models, without an understanding of human aspects, these technologies may struggle to align with user intentions and provide effective services. This perspective shifts the focus from “what artificial intelligence

should present to people” to “how people perceive artificial intelligence [19].”

Hence, the purpose of this research is to design experiments within a virtual collaboration environment to identify factors that influence human perception of personality and explore the potential of these factors to shape the perceived personality of AI. Specifically, this study focused on behavioral factors of AI as independent variables rather than linguistic elements and observed changes in participants’ behaviors and perceptions accordingly. These factors could be considered crucial aspects to consider in future AI personality research.

Chapter 2. Related Work

Licklider's prognostications concerning human-computer interaction (HCI) suggest that there is no singular approach to the use of tools by humans [20]. Additionally, the advent of computers has compelled us to redefine our relationship with pre-existing tools that humans have used for thousands of years. This perspective also holds true for HRI. By establishing a social connection with a robot or AI, individuals are no longer limited to conventional forms of interaction, perhaps even more so than with HCI more broadly. When employing a hammer or pencil, individuals typically do not attribute any personality or intentions to the tool; rather, they focus on its design or function.

However, when an entity exhibits more complex behaviors, it becomes difficult to perceive and judge the entity solely based on its outward appearance. Therefore, humans attempt to understand the entity's intentions and internal states. In doing so, subjective factors such as one's own values and behavioral patterns come into play alongside commonly accepted knowledge. Individuals make an effort to understand the other party by projecting their own perspectives and assumptions [21]. In other words, individuals interpret the robot as an entity with some manner of personality because of its behavior, not the other way around. The actual consciousness of the robot is irrelevant; what is significant is that individuals perceive the object as having intentions or desires based on observations of its behavior in-context.

Beyond that, the Eliza Effect is another phenomenon observed in HCI research, whereby individuals unconsciously anthropomorphize even the simplest forms of AI [15]. For example, in a study where a basic AI

algorithm was programmed to rephrase a patient's responses back to a question, individuals were found to accept the AI's behavior as similar to human behavior. Despite the algorithmic nature of the AI's responses, individuals still projected their own human-like qualities onto the AI and treated it as if it had a social presence. This complexity of human behavior provides clues for the development of better human-robot interaction (HRI), such as the development of AI individuals that attempt to trigger specific human mental positions, or the reflection of those positions within the AI's own behavior.

Human interaction with robots often involves perceiving them as entities with their own intentions and goals, irrespective of their actual capabilities. This phenomenon can be elucidated through Daniel Dennett's concept of the 'Intentional Stance' [22]. Dennett posits that when humans engage with their surroundings or any entity, they tend to ascribe intentions and desires to it, even in the absence of a mental system, personality, or physical needs. This perspective contrasts sharply with viewing an entity merely as a tool, characterized solely by its physical properties or design.

Interactions with robots that are limited to perceiving them as tools, governed by internal or external rules and devoid of self-motivation, fail to qualify as true interactive experiences, akin to using inanimate tools like hammers or screwdrivers. The complexity inherent in human behavioral stances towards robots offers valuable insights for the advancement of HRI. Specifically, it underscores the importance of developing AI personalities that not only evoke specific mental stances in humans but also mirror these stances in the AI's behavior. Such developments are pivotal in enhancing the depth and effectiveness of human-AI interactions (HAI).

Accordingly, even when dealing with pre-programmed robots that lack any assigned personality, humans still often tend to perceive them as having some sort of personality or goals [23]. This perception is not fixed and may differ depending on an individual's background, culture, and personality traits. Due to the complexity of the human social interaction, even the exact same output from a robot can be interpreted differently at different times, resulting in an unavoidable uncertainty in HRI from both a designer and user perspective.

Nevertheless, previous studies have demonstrated that adopting an intentional attitude towards non-human artifacts can enhance empathic responses and prosocial moral decision-making [24]. However, the fact that different individuals may react differently to the same response from a robot poses a potential risk of negative effects on the predictability of HRI in cooperative situations, which can, in turn, impact human trust in the robot [25].

On the other hand, the manipulation of speech speed and pitch to convey certain personalities in robots suggests that an artificially induced personality can guide human perception and reduce systematic uncertainty in communication during HRI [26]. By explicitly giving robots a personality, instead of relying on human imagination, we can create more accurate semantic communication that is better suited for a collaborative environment. This approach can help reduce HRI biases, misunderstandings, and uncertainties.

However, it is important to consider whether inducing personality in robots is simply a matter of mimicking human traits or if there are other factors that need to be considered. Failing to do so and relying on pre-programmed responses or hard-coded programming leads to a superficial

portrayal of personality that overlooks the intricacies and heterogeneity of human behavior and lacks a comprehensive understanding of the formation of a robot's personality. It would be nothing more than putting on a mask of "personality".

The concept of Inducing a personality in robots also raises ethical questions, particularly when it comes to mimicking human traits. This approach is not without its dangers, as evidenced by the "uncanny valley" effect, which has been reported in previous studies [27, 28]. The sense of otherness that arises when we recognize that a non-human being is only outwardly mimicking a human being only increases our discomfort with it, i.e., psychological distress. In addition, the lack of genuine human goals and intentions also contributes to this effect by producing cognitive dissonance [29].

These findings force us to reconsider the idea of manipulating not only the appearance but also the personality of robots in a mimetic way. It is essential to ensure that any induced personality in robots does not trigger negative emotional responses in humans. Therefore, before creating a personality in a robot, it is necessary to carefully consider the potential risks and benefits associated with such an approach. A more nuanced approach, one that considers the role of personality and individual differences in HRI, may be a more effective and ethical way to design robots that can interact with humans in a meaningful way.

In this study, a novel paradigm is adopted, diverging from the conventional approach of delineating rules dictating robotic behavior. Instead, the focus shifts towards understanding and learning how humans respond to robots [19]. This perspective conceptualizes personality not as a pre-defined and static attribute, but as a dynamic learning pattern forged

within the context of social interaction, thus providing behavioral stability. Such a viewpoint underscores the applicability of the “Intentional Stance” concept to robotics. It suggests that when humans perceive robots through an intentional stance, it necessitates those robots reciprocate by perceiving humans in a similar manner to facilitate natural social responses.

Consequently, this approach avoids basing the design of robots on established psychological frameworks such as the “Big 5” personality traits or the Myers-Brigg personality scale, which are limited in their applicability to robotic personality [30].

2-1. Intentional Stance in HRI

In the domain of HRI, numerous studies and experiments have been conducted based on the concept of the ‘Intentional Stance’. It is posited that humans naturally adopting an intentional stance is not only a result of evolutionary processes but also a pathway to more organic interactions [31]. Literature spanning philosophy, psychology, human development, and culture has been examined to suggest that adopting such a stance is crucial for achieving social collaboration with artificial agents. Furthermore, embracing this mindset is proposed as a pragmatic strategy enabling laypeople to comprehend and accept the behaviors of artificial agents. It is also argued that designing robots to exhibit such intentions could yield ethical outcomes [32].

However, it is important to acknowledge that humans do not uniformly perceive robots; individual variations exist in how robots are cognitively processed. To this end, a research team designed and evaluated the effectiveness of the Intentional Stance Task (IST) experiment to

measure the mental state activated during such interactions [33]. The study revealed that differences in the intentional stance taken by humans can manifest even when interacting with non-sentient entities, as evidenced by varied interpretations of the intentions behind identical robotic actions. Other recent literature also reports that the tendency to adopt an intentional or design stance towards robots may be influenced by individual differences [34], cognitive demands [35] expectations of the robot [36], as well as ingrained cultural values [37] and other factors.

A recent study by Meta AI in the context of a strategic diplomatic game, which necessitates collaboration, negotiation, and competition, integrates language models with reinforcement learning. This research developed an AI agent capable of conversing in natural language with humans while predicting and incorporating other's beliefs and intentions into its dialogue strategy [38]. Demonstrating human-level performance in actual play environments, the study underscores the potential for AI to adequately grasp human intentions through in-game data and dialogue. This suggests the feasibility of accessing the intentional stance of counterparts through environmental factors and conversational engagement with humans.

2-2. Personality in HRI

Along those lines, personality is already a significant topic in HRI, which is being investigated in two main directions: 1) robots perceiving human personality, and 2) humans perceiving robot personality. In both directions, a discernible, relatively stable personality is a crucial social factor for robots to respond to social interactions with humans because these traits are necessary to manifest consistent patterns of behavior,

thought, and emotion over the long-term, which allows for appropriate responses to be inferred efficiently, even in face-to-face situations with infinite possibilities [39]. For instance, if a person with bipolar disorder exhibits an erratic and unpredictable pattern of behavior, it can be very challenging for another person to cope with them.

Thus, this consistency contributes to a sense of stability in social interactions [40]. However, it should be noted that humans often consider themselves as being dehumanized if they adhere strictly to certain rules, and they accept a certain degree of irregularity as a natural and human trait. Therefore, when thinking about personality for robots, it cannot be approached in a simplified manner but rather requires careful consideration [41].

In a different vein of research, it has been found that individuals who have similar speech rates in the “Prisoner’s Dilemma” game are more likely to cooperate with each other, regardless of their level of liking each other. [42]. This implies that non-verbal factors (i.e., the way something is spoken), in addition to the speech content of communication (i.e., what is actually said), can influence cooperative behavior among people. As a result, investigating influential factors is critical for the development of cooperative social AI. If an individual’s optimal speech rate can impact the outcome of a cooperative task, then there is a need for systems that can manipulate these factors.

Similarly, another study aimed to investigate whether humans can identify a synthesized voice personality in text-to-speech (TTS) systems and its effect on engagement, trustworthiness, and perceived informativeness. The results indicated a similarity-attraction effect, where participants found the agent more engaging, trustworthy, and informative

when its parameters, such as words per minute or frequency range, matched their own unique speaking patterns in terms of known personality traits (introversion/extroversion) [36].

That finding supports the argument for the development of personalized AI with personalized personalities geared towards specific human users. However, conflicting results have been reported in other studies examining the effects of deliberately altering certain cues to explore how they affect 'people's perceptions and reactions to robots. These studies have shown that the more opposite an individual's personality was to that of the robot, the more intelligent and social it was perceived to be, known as the complementarity-attraction effect [34].

Therefore, the complexity and irregularity of human sub-conscious response mechanisms must be considered in any approach to how a robot appears to a person.

2-3. AI agents in virtual cooperative environments

Meanwhile, efforts have been made to implement social interaction between humans and artificial agents in virtual cooperative environments to explore such questions detailed above. For example, an experimental system was developed in which humans and AI interacted and cooperated within a video game environment called 'Don't Starve Together'. Using that system, a study was conducted in which participants were asked to perform a Turing Test, which involved determining whether the other individual was a human or an AI, based solely on the information in the virtual environment [43].

The findings showed that participants were less likely to identify the character they were playing with as an AI when it was an AI than when it was being controlled by a human. The follow-up study found that social AIs that exhibited active reactions to their opponents' actions in situations with limited verbal communication and no personality cues were perceived as more realistic than their human counterparts. In other words, people were surprisingly more likely to mistake the AI player they were interacting with for a human, compared to their human co-participant [44].

Furthermore, a previous experiment also employed real-time linguistic interaction between AI and humans within the same virtual environment [45]. What is noteworthy about this experiment is that it was conducted under various conditions using a speech system capable of making real-time utterances by considering the context through the state and changes in the environment, rather than a system that only responds to human linguistic signals. This demonstrates the methodological potential for exploring various factors in creating social AI.

Chapter 3. Experimental Design

For conducting the experiment, the elements include a virtual experimental environment where AI and participants can interact cooperatively, a virtual avatar to represent the AI's appearance, an AI system that enables the AI to operate autonomously in the experimental environment, and a speech system that allows voice interaction with participants. To construct the experimental environment, pre-existing programs were utilized, or systems developed directly in the laboratory were applied.

3-1. Virtual Cooperative Environment

The environment chosen for the interaction between AI characters and participants is a virtual cooperative game environment. This kind of virtual environment is emerging as a promising experimental space for rapidly collecting diverse data in the field of HRI. These tools offer the advantage of easily manipulating features of the environment to create different types of interactions while also efficiently capturing data that may be difficult to measure in the real world (or require specialized equipment to capture). They have the benefit of effectively collecting and observing human behavioral data. Through the existing speech system, participants can interact with AI via voice dialogue, with the content of the spoken dialogue directly connected to data about expressed behaviors of social entities (e.g., human and AI) but also situational elements in the environment surrounding them playing a crucial role.

Therefore, such virtual environments enable multifaceted data analysis approaches, providing the technical foundation necessary for

understanding the context of interactions. For the virtual cooperative simulation environment, the game "Don't Starve Together" (DST), developed by Klei Entertainment, was chosen. DST is a survival cooperative game where players must gather resources, craft tools, and produce necessary items and equipment to survive in a wild environment. Efficient task performance requires collaboration with other players, which thus necessitates social interaction. In this experiment, an AI that plays the game autonomously is deployed to collaborate with participants, while socially interacting with them via spoken dialogue at the same time.

Furthermore, DST is composed of game files in the Lua programming language, which supports user modification, allowing the editing and addition of arbitrary environments, scenarios, and Non-Playable Characters (NPCs). For these reasons, it has already been adopted as an experimental environment in other HRI experiments. To obtain experimental data in this virtual environment, a custom mod file (HRI_mod) was written by the researchers, which was capable of real-time collection of game state, AI behavior, and player behavior data.

Also, a user-defined environment for the experiments described below was preset, allowing experimenters to start each participant in an identical controlled scenario with the same map, location, and starting conditions. The experiments can store rich data generated in the game (via the game mod mentioned above), including the player's status, resources, tools, environmental elements, movement distance, motion, and current actions, in CSV files.

3-2. Virtual Avatar

Participants in the experiment engage not merely with in-game characters but experience an interaction akin to playing a real multiplayer game with a friend. This is facilitated through a Zoom meeting video call, where participants are face-to-face with a virtual 3D avatar. The aim is to elicit more natural and proactive social interactions from participants, rather than giving the impression of a program operating based on pre-entered commands. For the same reason, the AI is not intended as an “assistant” but rather a partner or collaborator in this scenario.

For the virtual avatar here, the program 'Loomie Live' developed by Loom.ai was utilized to create a life-like human avatar. The avatar possesses a specific appearance (i.e., we did not change the appearance for different participants during this experiment) and is capable of making gestures and lip movements corresponding to its speech. This approach enhances the engagement and realism of the experimental setup, fostering a more immersive and interactive experience for the participants.

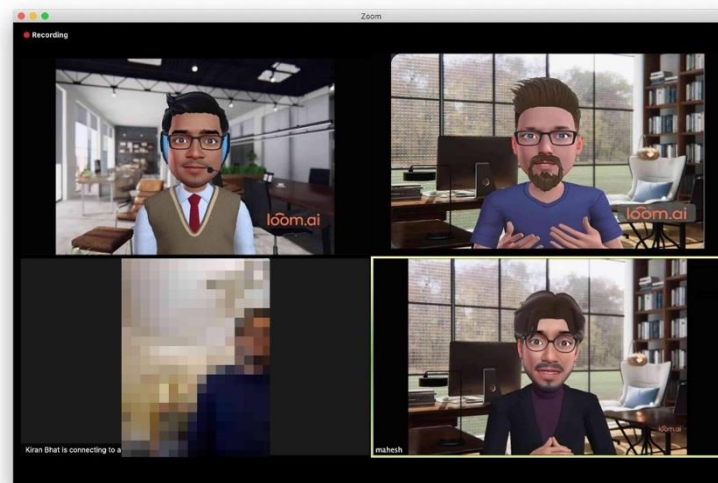


Figure 1. Example images for Loomie Live Avatar

3-3. Speech System

For the avatar to interact with participants through speech, it must automatically recognize the voice transmitted via microphone and convey appropriate responses through the speaker, tailored to the given utterance and context. To achieve this Automatic Speech Recognition (ASR), a Python program utilizing Microsoft Azure's speech-to-text API was developed. This program is also linked with the virtual avatar, enabling it to display corresponding mouth movements and motions as it speaks.

Although independent of DST, the avatar's responses are informed by a CSV file from the HRI_mod, reflecting in-game scenarios and the state of the participant or player. This system not only conveys its state vocally but also recognizes and responds to the participant's speech. It is connected to the AI system, controlling the in-game character to execute appropriate actions. For example, if a participant says, "Follow me," the avatar recognizes this command, decides to perform a 'Follow' action, and then controls the DST character accordingly. Simultaneously, it verbally responds through TTS, saying something like, "I'm coming to you."

The speech system used in the experiment, while in its initial build, does not generate generic utterances like ChatGPT. We have been working on a ChatGPT version of the speech system, but it was not finished at the time of these experiments. Instead, the speech system here categorizes the most frequently used words from the utterances of 60 Korean participants during previously conducted experiments in 2021–2022, using the KoNLPy Python Package. Those experiments included both human-AI interactions as well as human-human interactions during gameplay in DST. Using regular expressions, if a participant's utterance

includes any of those keywords, the system randomly selects one of several possible responses appropriate to the situation. For instance, if a participant says, "Bring me some wood," the system identifies this using a regular expression pattern like `".*(find|bring|fetch).*me."` and responds with one of the pre-scripted phrases like, 'Okay, I'll look for it,' 'This game map is so big,' or 'Sure, let's look for it together!'

3-4. Development of AI Personality System

To autonomously engage in gameplay and effectively implement distinct goals, action priorities, and behavioral patterns discernible among various agents, the following methodological approach was undertaken.

3-4-1. Autonomous AI System

The experiment leveraged the FAtiMA-Toolkit, an open-source architecture that provides a technical foundation for autonomous movement and actions of characters within DST. This toolkit, functioning as an appraisal and regulation process computational model, enables virtual agents to trigger in-game behaviors and simulate emotional responses to environmental events, including interactions with humans within the game. It addresses technical and conceptual challenges in agent-based tools, such as system heterogeneity, framework paradigms, real-time interaction complexities, and dialogue system structuring.

For this experiment, only a portion of the full toolkit was utilized, drawing upon and advancing a research project that generated Autonomous AI in DST (Figure 2). This AI System operates independently from DST, like the speech system, but perceives the game environment, situations, events,

and the states of itself and the player in real-time through HTTP communication with the game's backend. Based on that real-time data, the AI determines actions to execute, which are then communicated to a 2nd mod file (AI_mod) for the game. The AI_Mod is distinct from the HRI_mod above but runs in parallel. This 2nd mod file acts as an “AI puppeteer”, enabling the DST in-game character to perform fully autonomous actions without any human control. In this way, both the speech system and in-game character behavior are fully autonomous.

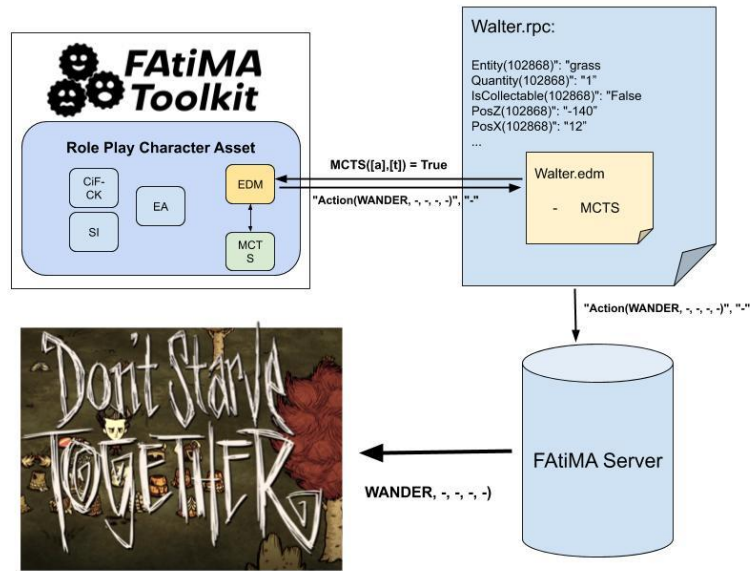


Figure 2. Schematic of DST AI system using FAtiMA-Toolkit

DST uses a “tick” method to track time during the game, which in the case of DST occurs roughly twice per second. Every tick, the AI system updates its awareness of the surrounding environment and its own state, exploring possible actions based on its beliefs. However, the AI system only provides general directives for actions to take, with fine-grained in-game movements for performing the action controlled by a Behavior Tree (BT) that is part of the core DST programming. Ultimately,

this AI system, integrated with the virtual avatar and speech system, offers participants an experience akin to playing the game with a human friend, but rather conversing with the avatar instead.

Additionally, the AI system is interconnected with the speech system, allowing it to react to the participant's utterances, not just based on in-game situations or its state. For instance, if a participant says, "Come here, please." the AI not only responds with a matching utterance, like "Okay, I'm coming to you," but can also trigger response behaviors by the in-game character (e.g. moving towards the participant). This synchronization fosters more natural social interactions.

3-4-2. AI Personality

Currently, AI systems possessing human-level consciousness or self-awareness do not yet exist, making the implementation of genuine personalities in AI unfeasible. However, we propose a system that offers a novel perspective on AI personality, considering human perception modalities. This approach aims to develop an AI personality that is explainable based on data as conceived by its designers. **In other words, a data-driven approach to AI personality development.** To experiment with this concept, we developed AI personality system that exhibits varying behavioral patterns based on parameter adjustments.

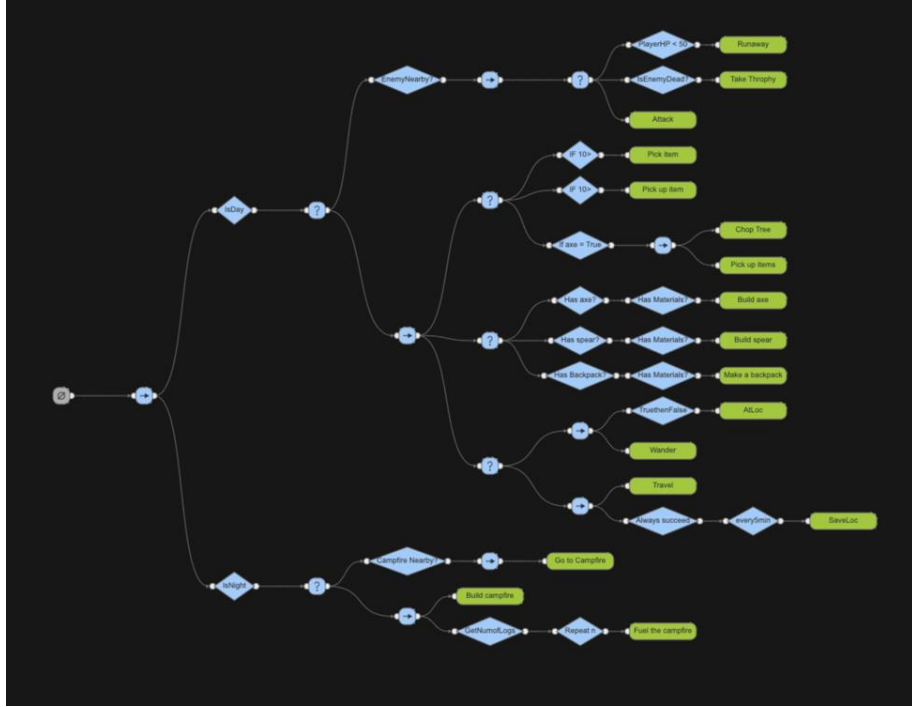


Figure 3. Behavior Tree of Adventurer Personality

This system implements distinct behavioral patterns using various BTs in the FATiMA-toolkit-based Autonomous AI system. BTs used in game AI development, featuring nodes like Sequence, Selector, and Loop, are interconnected in logical structures analogous to AND, OR, and IF conditions as shown in Figure 3. They perform depth-first searches to yield final outcomes. By setting goals and priorities for the AI, it can take different actions in the same situation depending on the tree's structure. Each action is governed by specific conditions and priorities, ultimately selecting one to command the DST character's movements. For instance, one agent might be programmed to explore new environments actively, while another might be restricted to roaming around the camp. Additionally, the AI can be controlled to decide whether to finish its current task before moving closer

to a participant who is too far away, or to immediately stop and approach once a certain distance is exceeded.

In this experimental endeavor, three externally manifestable personalities were proposed: *Adventurer*, *Camper*, and *Supporter*. Each personality archetype embodies distinct objectives and priorities, discernible through 'measurable' in-game goals. The behavioral expressions of each personality are characterized as follows:

- Adventurer: Primarily exploration-oriented, evaluated by a vision score.
- Camper: Primarily focused on resource gathering, gauged by a resource score.
- Supporter: Focused on player behavior, evaluated by a Distance score.

To operationalize these personalities within the AI system, parameters were configured to modulate AI behaviors. These parameters represent independent behavioral conditions, not explicitly tied to any specific personality. However, as evidenced in the subsequent pretests below, the modulation of these parameters influenced how utilization of the BT by the AI, thus manifesting distinguishable data patterns aligned with the proposed personalities (Figure 4). The parameters were fine-tuned through trial-and-error to exhibit the desired distinct traits for each personality. The finalized parameters employed in the experiment are detailed in Table 1.

Table 1. Parameter settings for AI personality

	No Personality	Adventurer	Camper	Supporter
max_player_dist	1024	4096	1024	256
max_wander_dist	Math.huge	Math.huge	4096	Math.huge
pick_chance	0.5	0.2	1	1
attack_chance	0.5	1	0.5	0.5
travel_chance	0	0.5	0	0
gohome_chance	0	0	0.8	0

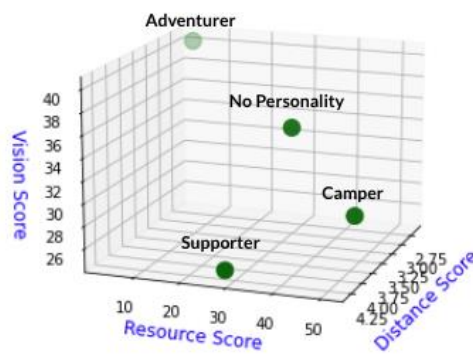


Figure 4. Each personality's metric scores on the pretest

Through these mechanisms, the AI personality system not only adapts to varied scenarios but also aligns with participant interactions, contributing to a more dynamic and engaging gaming experience.

3-5. Pretest

Prior to the actual experiment, a pretest involving five laboratory personnel was conducted to develop the optimal parameters for each AI personality while also ensuring smooth execution of later participant experiments. To set up the experimental environment, two separate computers were prepared: one for the participant to play the game with the Avatar (player computer) and another where the Virtual Avatar exists in a distinct space (avatar computer).

The player and avatar computers were in separate spaces. Participants used the player computer, equipped with a camera to record their facial expressions and a microphone for voice communication. The audio signals were transmitted to the Avatar Computer via Zoom meeting.

On the Avatar Computer, the Speech System, AI Personality System, Zoom meeting, and the DST game were all run simultaneously to check if data was being recorded and processed correctly. Hosting of the private multiplayer session of Don't Starve Together was also conducted here, while the player computer simply accessed this created space. The gameplay and participant's facial videos from the Zoom Meeting were recorded on this computer using OBS Studio. After the pretest, bugs and errors identified during the process were rectified.

3-6. Participants

Following the pretest, a recruitment announcement was made for face-to-face experiment participants, targeting individuals over 18 years of age who were proficient in Korean. To ensure a generalized sample, high levels of gaming experience or skill were not required from the participants, though each participant was provided a brief “tutorial” of DST gameplay prior to the experiment. A total of 36 Korean participants were recruited, with 12 participants for each of the three personality types ($M_{age} = 23.6$, $SD = 2.3$, $min = 20$, $max = 29$, comprising 18 males and 18 females). All experimental procedures and personal information handling were conducted in accordance with the Institutional Review Board (IRB) guidelines of Hanyang University.

3-7. Experimental Procedures

Participants visited the experimental site according to a predetermined schedule. Upon arrival, they signed a consent form for the use of personal information, following the researcher's guidance. They were then briefed about the study and introduced to the basics of the DST game, including fundamental information and gameplay controls. This briefing helped participants understand the actions necessary for survival in the wilderness, how to procure food, and craft tools. They were also informed about the avatar waiting in the Zoom meeting and instructed on how to communicate with it using a headset and microphone to collaboratively play the game.

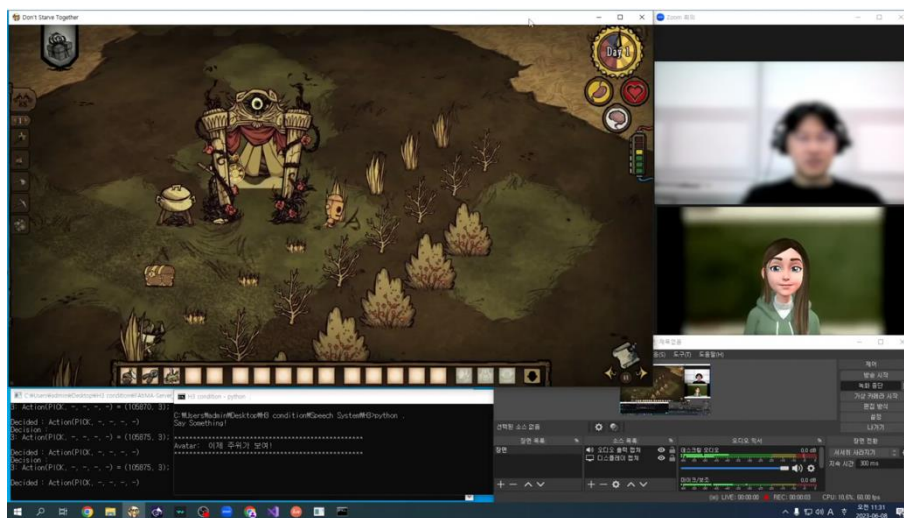


Figure 5. Screen capture of an avatar computer during an experiment

Next, participants cooperated with an AI possessing a pre-set AI Personality for a 30-minute game session. All interactions during gameplay were conducted in Korean. Participants could ask questions of the Avatar or request resources and food. While researchers continuously monitored the progress from the Avatar Computer, intervening in cases of program

errors or unforeseen circumstances, interference in the experiment was kept to a minimum. Upon completion of the gameplay, participants were asked to complete a survey. The survey consisted of both open-ended and closed-ended questions. The closed-ended portion included standardized HRI assessment instruments: the Godspeed scale (Bartneck et al., 2009) for measuring general perceptions of a robot/agent, and the Networked Minds instrument (Biocca et al., 2001; Oh et al., 2018) for measuring social presence. The open-ended section involved descriptive questions from a specially prepared Avatar Interaction Questionnaire, to gather qualitative data about participant's experience with the AI.

Chapter 4. Results

4-1. Data Description

The data obtained through the experiment is categorized into three distinct types: 'In-game Behavior Data,' which encompasses the gaming activity logs of both players and avatars within the DST, 'Speech Script Data,' representing the records of interactions between participants conducted via Zoom; and 'Instruments Data,' which consists of information provided by participants after the experiment has concluded. Each of these data types was consolidated and managed using the participants' unique experiment IDs.

4-1-1. In-game Behavior Data

In this environment, the collected data features are as follows, presented in the Table 2.

Table 2. Description of Data Features collected in DST

Feature Name	Description	Data Type
OS_timestamp	Time on the system	float
Game_Runtime	Time spent in game	float
Phase	day/evening/night	categorical
ID	ID of Avatar / PLAYER	string
Xloc	X Coordinate in Game	float
Yloc	Y Coordinate in Game	float
Zloc	Z Coordinate in Game	float
Curr_Inv_Cnt	Number of items in the current inventory	int
Curr_Active_Item	Id of currently active item	string

Curr_Equip_Hands	Id of Item currently in-hand	string
Health	Current Health Points	int
Hunger	Current Hunger Points	int
Sanity	Current Sanity Points	int
Attack_Target	ID of the target currently attacking	string
Defense_Target	ID of the target currently defense	string
Recent_attacked	ID of recently attacked targets	string
Food_value(hunger)	How much food can be filled with the food in the inventory	float
Twigs	Number of twigs in the inventory	int
Flint	Number of flints in the inventory	int
Rock	Number of rocks in the inventory	int
Grass	Number of grasses in the inventory	int
Log	Number of logs in the inventory	int
Tool_resource	Materials used to build tools (twig:flint)	string
Fireplace_resource	Materials Used to Craft Fire (logs:rock:grass:flint)	string
Is_Fireplace	Whether it's near a fire	boolean
Is_Light	Whether it's near a light source	boolean
Is_Monster(num)	Number of monsters nearby	int
Action	The action currently being performed	string
Target	The target ID of current action	string
Invobject	Tools related to the current action	string
IsRunning	Whether it's moving or not	boolean
Rotation	Currently facing angle	float
Dist_from_Basecamp	Distance from start point	float
Distance	Distance between Avatar and Player	float

4-1-2. Speech Script Data

During the experiment, the participants', and avatars' conversation records are documented and stored as shown in Table 3. Participants are distinguished and denoted by '1,' while avatars are indicated by '2.' The 'Time range' column indicates the start and end times of each utterance, and 'Utterance' contains the actual content of the conversation.

Table 3. Sample of Speech Script record

Speaker	Time	Utterance	Translation (ENG)
1	02:52 : 02:53	토끼 잡으러 가자	Let's catch a rabbit
2	03:42 : 03:44	그래 그건 어디 있어	Okay, where's that?
1	03:47 : 03:48	안쪽으로 가자	Let's go inside
2	03:50 : 03:51	길을 모르겠어	I can't find my way
1	03:55 : 03:57	얼른 따라와 너 어디 갔어	Come on, where did you go?
2	03:59 : 04:01	혹시 도움이 필요한 거야?	Do you need any help?
1	04:02 : 04:05	아니 저기 아무것도 없으니까. 안으로 가보자구	No, there's nothing there. Let's go inside.

furthermore, utilizing the recorded conversation data from each experiment, statistical measures concerning the number of utterances by both human participants and avatars within each experimental condition were computed, as depicted in Table 4.

Table 4. Descriptive statistics of Utterance Counts

	Human			Avatar		
AI Personality	Adventurer	Camper	Supporter	Adventurer	Camper	Supporter
Mean	60.75	62.83	49.33	48.75	50.75	48.83
Standard Error	9.04	6.65	4.83	4.89	5.55	4.32
Median	69.5	63	47.5	47	45.5	43.5
Mode	72	63	45	53	68	39
SD	31.34	23.04	16.74	16.96	19.23	14.98
Sample Variance	982.38	530.87	280.42	287.84	370.02	224.51
Kurtosis	-0.6708	0.3412	-0.3499	0.1535	-1.36	-1.37
Skewness	-0.3881	0.3048	0.5885	0.7195	0.3457	0.4193
Range	97	82	53	55	55	44
Minimum	12	24	28	29	25	30
Maximum	109	106	81	84	80	74
Sum	729	754	592	585	609	586
Count	12	12	12	12	12	12

Based on the data, an analysis of variance (ANOVA) was conducted to investigate whether the distribution of player utterance counts varies significantly across different AI personalities. However, the results indicate that there was no statistically significant difference observed among the groups, as evidenced by a p-value of 0.435.

4-1-3. Instruments Data

Through the Godspeed questionnaire, participants can express their impressions of the robot using scores ranging from 1 to 5 points. The domains for assessment include 'Anthropomorphism,' 'Animacy,' 'Likeability', 'Perceived Intelligence', and 'Perceived Safety.' These domains comprise specific items such as 'Fake' vs. 'Natural', 'Apathetic' vs. 'Responsive', and others.

Meanwhile, the Networked Minds instrument comprises two distinct components: 'self' and 'other.' The 'self' component focuses on the assessment of individuals' self-perceptions, while the 'other' component involves evaluating individuals' perceptions of the other party. Within each component, participants provide responses related to specific domains, including 'Co-Presence,' 'Psycho-Behavioral Interaction,' 'Perceived Attentional Engagement,' 'Perceived Emotional Contagion,' 'Perceived Comprehension,' and 'Behavioral Interdependence'.

For instance, within the 'Self' section, the questionnaire for 'Co-Presence' is structured as follows, and similar to Godspeed, it is evaluated using scores ranging from 1 to 5 points:

- 'I often felt as if the avatar and I were in the same game together.'
- 'I was often aware of the avatar in the game.'
- 'I hardly noticed the avatar in the game. '

Participants provide ratings for these statements to indicate their level of agreement or perception regarding the sense of co-presence in the virtual environment.

After the experiments, the One-way ANOVA analysis was conducted on the Godspeed questionnaire data as well. However, it was challenging to observe significant changes in participant responses based on AI Personality across all six domains. Particularly, the averages and standard deviations for each personality group in the Godspeed questionnaire showed minimal differences: Adventurer (Mean, 2.98, SD, 0.513), Camper (Mean, 2.88, SD, 0.478), and Supporter (Mean, 3.04, SD, 0.25).

Similarly, ANOVA analyses were conducted on the Networked Minds questionnaire data, but neither questionnaire revealed dramatic distribution differences across all seven domains. For Networked Minds (Self), the overall means and standard deviations were as follows: Adventurer (Mean, 2.98, SD, 0.378), Camper (Mean, 2.91, SD, 0.223), and Supporter (Mean, 3.12, SD, 0.256). In Networked Minds (Other), the means and standard deviations were Adventurer (Mean, 3.21, SD, 0.4), Camper (Mean, 3.06, SD, 0.274), and Supporter (Mean, 3.28, SD, 0.283).

Regarding the ANOVA results, for Networked Minds (Self), all domains, except for 'Perceived Emotional Contagion,' had p-values greater than 0.05, with 'Perceived Emotional Contagion' showing a significant difference ($p = .038$). In Networked Minds (Other), all domains, except for 'Behavioral Inter-dependence,' had p-values greater than 0.05, with 'Behavioral Inter-dependence' indicating a significant difference ($p = .025$).

4-2. Data Analysis

This section describes the results of analyzing the data obtained from the experiment. Compare and examine the data distribution, proportion, etc. according to each condition.

4-2-1. Player Actions

As mentioned in Section 4.1.1, it is possible to record all actions performed by participants during the experiment. The graph representing the proportion of actions that occurred in each experimental condition is depicted in **Figure 6, 7**. Across all experimental conditions, it is evident that the top 5 actions (PICK, ATTACK, CHOP, PICKUP, EAT) were the most frequently performed. This indicates that these common actions, such as cutting trees, hunting animals, and gathering resources from the ground, played a predominant role in survival.

However, when examining the graphs for participants who played alongside the Adventurer AI and those who played with the Camper AI, differences in the order of the top 5 actions can be observed. The former group exhibited a higher proportion of ATTACK, while the latter group showed higher proportions of CHOP and PICKUP.

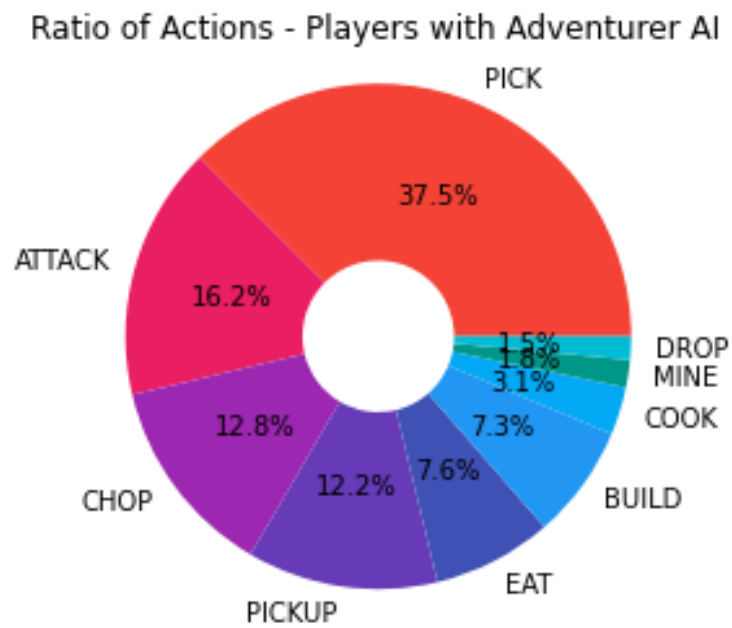


Figure 6. Percentage of Player's Actions played with Adventurer AI

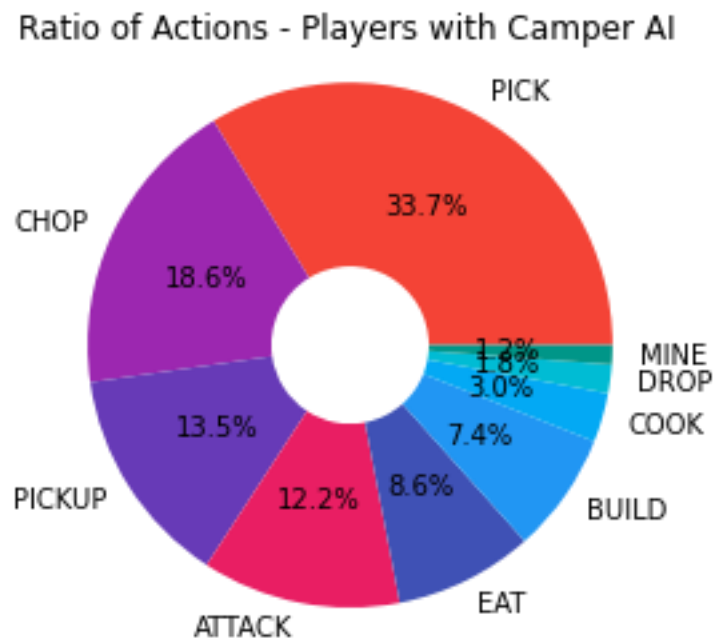


Figure 7. Percentage of Player's Actions played with Camper AI

4-2-2. Player Movements

In Section 3-4-2, measurable metric scores were defined to implement distinct behaviors for each AI personality. For example, the Camper AI, whose primary objective is resource collection, was configured to have a high Resource Score based on the quantity of resources collected. On the other hand, the Supporter AI was assigned a higher Distance Score when it was closer to the player during the experiment. It is worth noting that the Distance Score was computed by taking the negative logarithm.

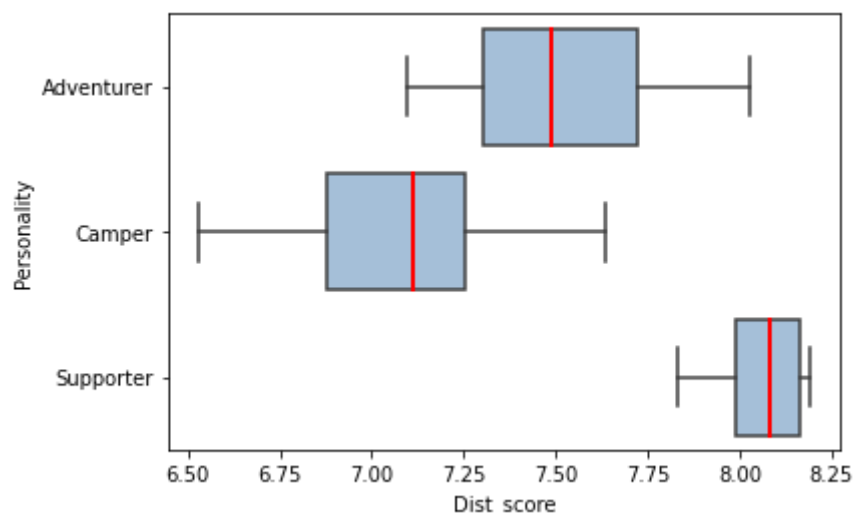


Figure 8. Box plots showing that distance according to each personality.

Figure 8. revealed that the Distance Scores of participants varied significantly across different conditions ($p < .001$). However, this outcome is indicative of the results driven by AI behavior patterns, and it does not necessarily reflect how differently participants behaved during the actual gameplay.

The distance of the player from the starting point, known as Basecamp, during the game could be measured using the

'Dist_from_Basecamp' feature. Basecamp is equipped with various tools and tents that are advantageous for survival. In practice, participants referred to this location as 'Home' and tended to return to it when they were far away and as night fell.

However, as shown in Figure 9, there was no significant difference in the tendency of participants across different groups to stay close to the starting point and prefer the immediate vicinity ($p = .233$). Interestingly, as indicated in Table 1, despite the AI personalities being designed to encourage AI to move closer or farther from the basecamp, each group exhibited a wide variance in their behavior.

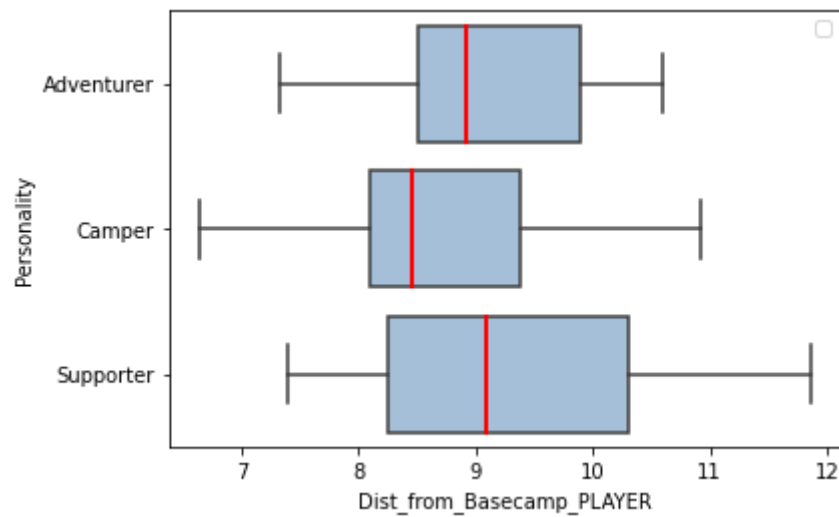


Figure 9. Box plots showing that average distance from basecamp of players.

On the other hand, viewing the participants' movement paths not in terms of distance but as a single pattern enables the assessment of how 'random' their movement paths were. For computational convenience, the areas where the player moved were divided into grids, and the frequency of their visits to each grid was represented as a probability. As shown in Figure 10, this approach allows for the application of Shannon's Entropy

formula, depicting the heatmap of the player's movement path.

This difference was more readily apparent when examining only the heatmap (Figure 11). Participant KOR77 remained near the (0, 0) coordinates, indicating their proximity to the starting point, the base camp. In contrast, participant KOR68 actively ventured far from the starting point and exhibited a more complex movement pattern, deviating from predictable routes.

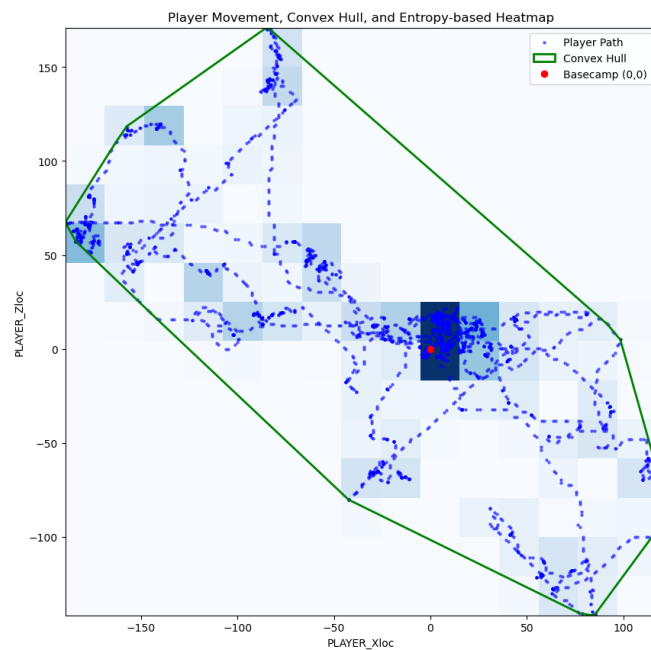


Figure 10. Representation of a participant's entire travel path

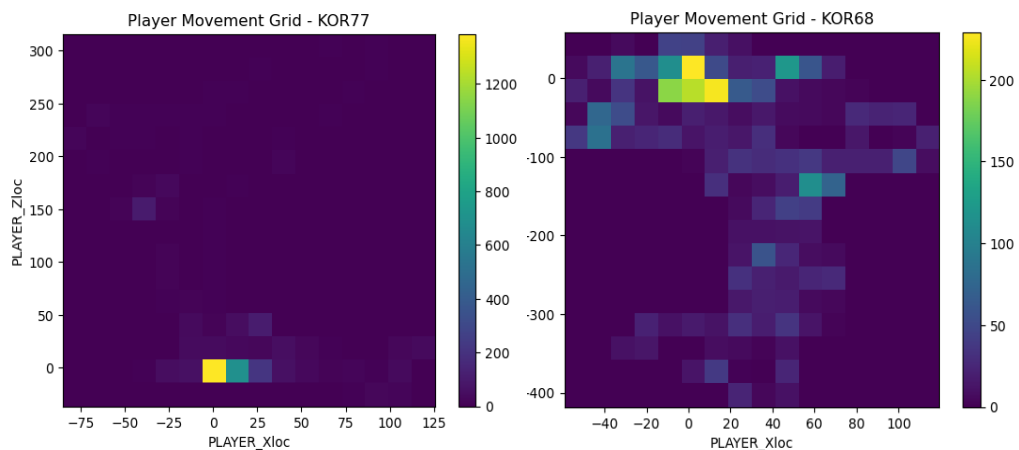


Figure 11. Two different participants' travel paths Heatmap

When each player's movement path was divided into grids and the entropy formula was applied, a correlation coefficient value of -0.5024 was obtained in relation to the “Proximity” that reflects how close players were to AI during the experiment (Figure 12). This correlation suggests that as the distance between participants and AI was maintained closely during the experiment, the player's movements became less random. Moreover, when considering which AI personality, the participants interacted with, it was observed that each group exhibited distinct distributions. However, it's important to note that this represents a correlation, not causation, and further detailed analysis is needed to determine the actual impact.

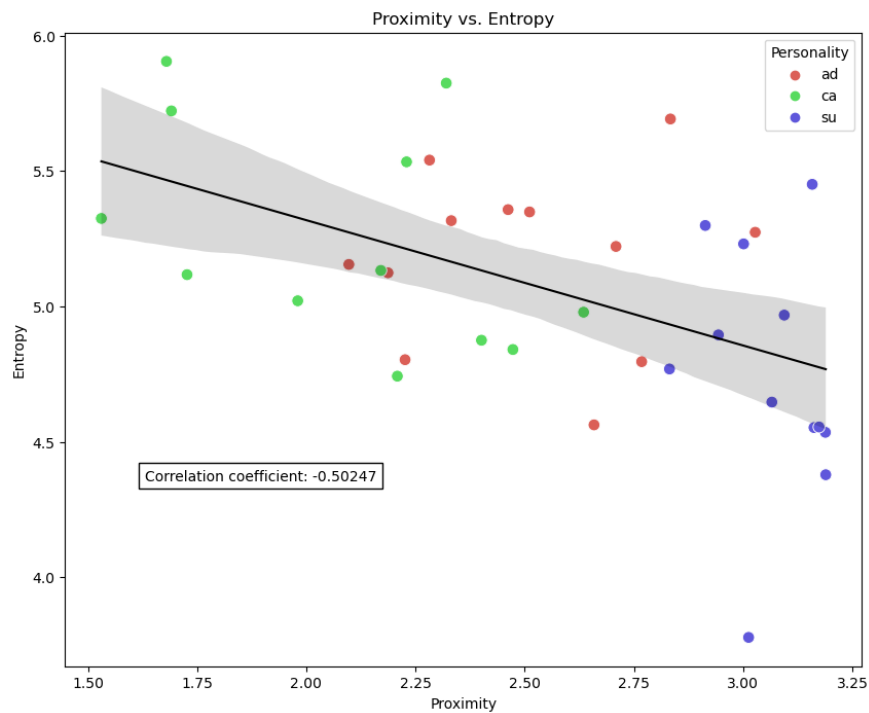


Figure 12. Correlation between Proximity and Entropy

4-3. Modeling

During the analysis process, ANOVA allowed for the examination of differences in specific features among different experimental condition groups. However, to gain a comprehensive understanding of how humans perceive AI Personality, exploring integrated information rather than individual features becomes necessary. Hence, modeling was conducted to classify and predict AI personality using time-series data, specifically the In-game data.

However, the dataset consists of a multiclass classification problem with three labels: 'Adventurer,' 'Camper,' and 'Supporter'. Since there are only a total of 36 samples (12 for each class), dividing them into training and testing sets would result in very small training sets, making evaluation challenging. An attempt was made to employ Data Augmentation using the SMOTE technique, which is one of the methods in the domain of Few-Shot Learning (FSL). SMOTE_OUT was utilized to expand the dataset for each distinct label category. However, due to the persistently insufficient number of samples in comparison to the intricacies of the dataset (e.g., the large number of features/variables), noteworthy outcomes were not attained showing accuracy 45%, AUC = 0.4081.

Chapter 5. Discussion

5-1. Summary of Results

This study focused on designing experiments within a virtual collaboration environment to identify factors influencing human reactions to AI personalities as an extension of existing research [45]. By examining various data sources, including 'In-game behavior data' representing activities in the virtual environment, 'Speech script data' recording conversations between avatars and participants, and post-experiment survey data on AI impressions and feelings, we conducted ANOVA analyses to validate several factors related to human reactions and behaviors based on AI's personality.

Regarding 'In-game behavior data,' while multiple features were available, our analysis primarily focused on participant actions and movements. Particularly, in the aspect of the distance between participants and avatars, a trend was observed where reducing the distance between them led to a decrease in the randomness of participants' movement patterns.

Analysis of 'Speech script data' involved examining the number of utterances during conversations. The results did not show significant differences between groups based on AI personalities. This was somewhat expected, as all three experimental conditions used the same speech system. Additionally, there were significant variations in the frequency of utterances among participants, highlighting differences between talkative and reserved individuals.

'Instruments Data' did not yield significant differences between the experimental groups either. When examining participants' responses in the

Godspeed and Networked Mind surveys, most evaluations clustered around the 3-point range, making it challenging to discern differences between groups.

In the end, utilizing classification models based on participants' behavioral data, an attempt was made to predict the personality of the AI with which participants interacted. However, due to the limited number of samples in each class, even with the addition of data through SMOTE, high accuracy could not be achieved.

As a result, contrary to expectations, this experiment did not find substantial changes in human behavior based on changes in AI behavior. Several anticipated factors may contribute to these outcomes.

Firstly, the simultaneous execution of tasks and interactions posed a significant challenge to participants. They were provided with basic instructions about the operation of DST and the environment before the experiment. However, exposure to an unfamiliar environment within the initial 30 minutes made it difficult for participants to leisurely explore their surroundings, even when the Avatar was nearby. Instances were observed where participants perceived themselves as distant from the Avatar, or they spent considerable time wandering while attempting to obtain specific items. While a reasonable level of difficulty can serve as a motivating factor for participants to engage with the Avatar and seek assistance or ask questions, excessively high levels of difficulty may unintentionally hinder interaction [46]. Therefore, it may be advisable to consider creating a simpler or more straightforward environment in future experiments.

Secondly, there is a need for more pronounced changes in Avatar behavior patterns based on conditions. These changes should extend beyond measurable metrics and be perceivable by human awareness as

entirely distinct patterns [47]. Currently, the experiment primarily relies on parameters such as the probability of attempting specific actions and distances from particular objects. The application of additional elements is necessary. This issue is also interconnected with the necessity to enhance the diversity of AI behavior.

Lastly, it is worth noting that the influence of AI behavior on humans may be subconscious. Research examining collaborative task execution between AI and humans suggests that both verbal and non-verbal communication play a crucial role in reinforcing the imagined "We," signifying an implicit agreement among group members with different entities cooperating towards a shared goal [48]. Here, "We" represents the concept of an implicit agreement among different entities working towards a common objective. These findings suggest that humans may unconsciously synchronize their behavior with other collaborators due to the existence of mechanisms that facilitate such synchronization. Changes in participant entropy during this experiment did not strongly correlate with responses in the Godspeed and Networked Mind surveys. Therefore, alternative methods may be required to measure these aspects effectively.

5-2. Limitations

One significant limitation of the largest experiment is the fact that AI's behavior is still not as natural as a rule-based system. It demonstrated limitations in displaying as diverse behavioral patterns as those exhibited by humans. However, employing overly complex Behavior Trees can also pose challenges in controlling AI behavior. Instead, considering reinforcement learning agents that act with different goals in mind could be

more suitable for this type of experiment. These agents can exhibit various behavioral patterns without explicit rules, making them more adaptable to the experiment's goals.

Regarding the speech system, it's worth noting that systems like ChatGPT generate speech by randomly selecting text from a predefined set, rather than generating responses dynamically. Consequently, it often fails to provide accurate responses to participant's questions or requests, leading to unnatural interactions. This could be addressed by implementing conversation generation methods using LLMs, which do not require predicting all possible responses in advance.

Furthermore, a significant challenge faced during the experiment was the variance in participant's' experience and immersion [49]. Especially in the DST condition, while it offers a clear survival goal, it does not provide specific detailed objectives. This resulted in some participants struggling to determine what they should do, particularly in the open-world scenario. Conversely, participants with prior DST game experience tended to proceed independently without relying on AI assistance.

Finally, due to the limited number of samples per class, the classification models did not yield significant predictive results. Conducting face-to-face experiments, where each participant's session took approximately an hour, posed constraints on the number of experiments that could be conducted. Transitioning to an online format would potentially enable a broader range of participants to engage in the experiments more frequently.

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국문 요지

최근 기술의 발전으로 인공지능은 인간의 단순한 보조 역할이 아닌 서로 간의 상호작용이 일어나는 협업 중심의 기술이 되었다. 이때 AI의 일관된 행동 패턴과 특징의 적용은 실제 환경에서 상호작용의 불확실성을 줄일 수 있게 되지만 이것은 '명시적' 접근 방식을 통해 AI 기능을 조작하는 것에 불과하다. 또한 이와 별개로 인간은 스스로 AI를 의인화하고, 마치 성격이 있는 것처럼 인식하는 경향이 있어 둘 사이의 충돌이 일어날 수 있는 문제가 있다. 따라서 이 연구에서는 AI 성격에 대한 인간의 행동적 변화를 형성하는 요인을 탐색하기 위해 각각의 개성을 가진 AI와 협동하여 생존해야 하는 가상 환경에서의 실험을 진행하였다. 이를 통해 향후 AI 성격 연구를 위한 고려 사항과 요인들을 조명할 수 있다. 총 36명의 한국인 실험 참가자는 'Don't Starve Together' 게임에서 실시간 음성 대화로 상호작용할 수 있는 AI Agent와 30분간 같이 플레이를 하였다. 실험 후, 참가자에 대한 행동, 발화, 설문 데이터를 수집하였으며 각 실험군이 통계적으로 유의한 차이를 보여주는지 비교하였다. 분석 결과 참가자의 이동 경로에 대한 Entropy가 집단 간 유의한 차이를 보여주었다. 하지만 같이 플레이한 AI에 따른 발화 수나, 설문에서는 유의한 결과가 나타나지 않았다. 이후 참가자의 행동 데이터를 이용하여 어떤 AI 개성과 플레이 했는지 예측하는 모델을 구축하고자 시도했으나 학습 및 데이터 세트의 부족으로 인해 충분한 분류 성능은 보여주지 못했다. 이후 연구에서는 강화 학습을 통한 Agent와 생성 모델을 적용한 Speech System을 통해 기계적이지 않고 자연스러운 상호작용을 할 수 있는 실험을 시도해볼 수 있다.

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Declaration of Ethical Conduct in Research

I, as a graduate student of Hanyang University, hereby declare that I have abided by the following Code of Research Ethics while writing this dissertation thesis, during my degree program.

"First, I have strived to be honest in my conduct, to produce valid and reliable research conforming with the guidance of my thesis supervisor, and I affirm that my thesis contains honest, fair and reasonable conclusions based on my own careful research under the guidance of my thesis supervisor.

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DECEMBER 19, 2023

Degree : Master
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Thesis Supervisor : Casey Bennett
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연구 윤리 서약서

본인은 한양대학교 대학원생으로서 이 학위논문 작성 과정에서 다음과 같이 연구 윤리의 기본 원칙을 준수하였음을 서약합니다.

첫째, 지도교수의 지도를 받아 정직하고 엄정한 연구를 수행하여 학위논문을 작성한다.

둘째, 논문 작성시 위조, 변조, 표절 등 학문적 진실성을 훼손하는 어떤 연구 부정행위도 하지 않는다.

셋째, 논문 작성시 논문유사도 검증시스템 "카피킬러"등을 거쳐야 한다.

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